Name: SOLUTIONS Date: ______

# Year 9 Mathematics

## Test 5, 2015

### Topics – Statistics

| |
|---|
| ___ |
| 55 |
| = ___ % |

**Total Time:** *60 minutes*

**Total Reading:** *5 minutes*

**Total Working:** *55 minutes*

**Weighting:** ______*% of the year.*

## SECTION 1: CALCULATOR FREE

| | | | |
|---|---|---|---|
| **Time:** | *23 minutes* | **Marks for Section 1:** | *21* |
| **Reading:** | *2 minutes* | **Equipment Allowed:** | *Nil* |
| **Working:** | *21 minutes* | | |

**1. [2 marks: ½ each]**

| | | | |
|---|---|---|---|
| *categorical data* | *range* | *continuous data* | *discrete data* |
| *mean* | *skewed* | *symmetrical* | *mode* |

Complete the following using words from the list above.

a) The number of goals scored by each team in one round of a netball competition is an example of

discrete data ½

b) If a statistical graph is not skewed it is said to be symmetrical. ½.

c) The height of each student in your class is an example of continuous. data ½

d) The difference between the highest and lowest score is the range ½.

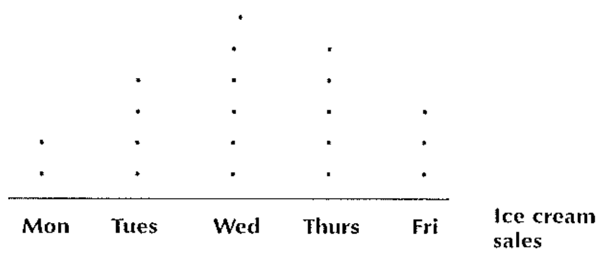
**2. [2 marks]**
Explain why your weight is considered to be continuous data rather than discrete even though you record it as 52 kg, 55 kg, etc.

continuous because every value between 52 kg and 53 kg, for example, can be represented at some point ✓

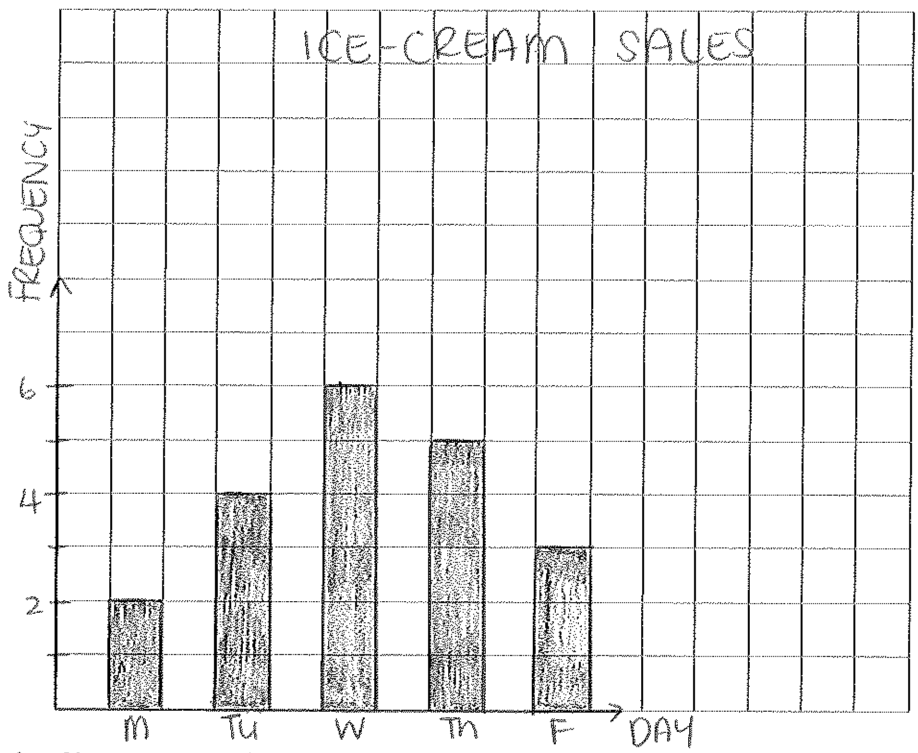
we round because it is easier to record ✓

or other reasonable explanation (max 2).

**3.** **[5 mark: 2, 1, 1, 1]**
The sale of ice creams from the school canteen was recorded for the past 15 days.




**a)** Display this information as a column graph.




✓labels
✓columns
$-\frac{1}{2}$ no title
$-\frac{1}{2}$ no gaps.

**b)** When did the sale of ice creams peak?

Wednesday ✓

**c)** What was the average number of ice creams?

$\frac{20}{5} = 4$ ✓

**d)** What day do you think was the coldest? Give reasons for your answer.

Monday $\frac{1}{2}$
least number of ice-creams bought $\frac{1}{2}$.

**4.** **[1 mark]**
Abbey scores the following scores in 6 hands of a card game.
200, 180, 160, 140, 100 and 40.

Find the median of her scores. $\frac{140+160}{2} = 150$ ✓

**5.** **[2 marks]**
Five friends earn the following marks on a quiz out of 10 marks:

5, 5, 6, 9 and 10.

Find the mean of their marks.

$$\frac{5+5+6+9+10}{5} = \frac{35}{5} = 7$$ ✓

**6.** **[1 mark]**
Ellie plays eight games of netball and scores the following number of goals:

8, 3, 6, 7, 12, 9 and 14.

What is the range of her scores? $14 - 3 = 11$ ✓

**7.** **[1 mark]**
Taylah records the number of minutes of exercise she does each day over two weeks.

| | Mon | Tue | Wed | Thur | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| Week 1 | 20 | 45 | 30 | 20 | 60 | 45 | 60 |
| Week 2 | 45 | 20 | 45 | 30 | 20 | 45 | 35 |

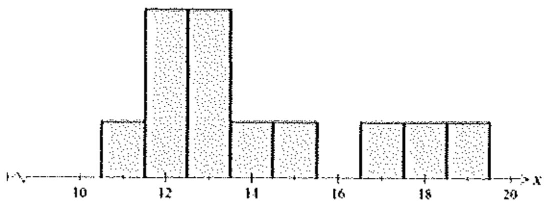
20 - 4
45 - 5

Find the mode of her times.

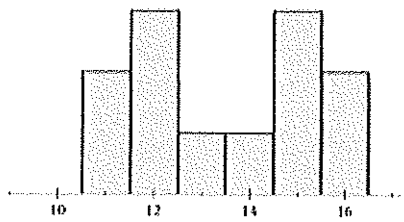
45 min ✓

**8.** **[1 mark]**
Which histogram could **not** be described as being bimodal?

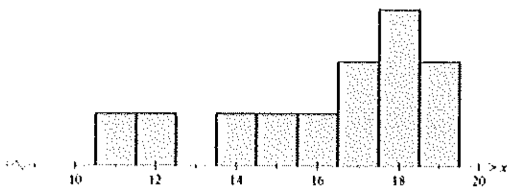
**A**



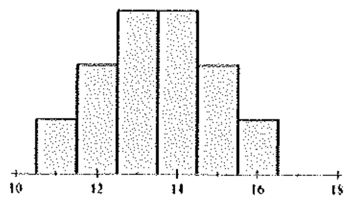

**B**




**C**




**D**




✓

**9. [3 marks: 1, 1, 1]**

Mitchell's scores in 20 shots at a dartboard.

| Stem | Leaf | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 3 | 5 | 7 | 9 | 9 | | | |
| 1 | 0 | 4 | 7 | 8 | 9 | 9 | | |
| 2 | 2 | 5 | 6 | 6 | 6 | 6 | 9 | 9 |
| 3 | | | | | | | | |
| 4 | | | | | | | | |
| 5 | | | | | | | | |
| 6 | 0 | | | | | | | |

**a)** What is the mode of the scores?

26 ✓

**b)** Which score could be described as an outlier?

60 ✓

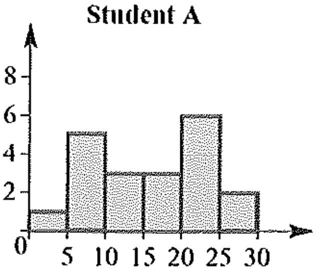
**c)** What effect would it have on the mean if the outlier were not included in the data? (You do not have to calculate the mean to answer this question.)

mean would be reduced ✓

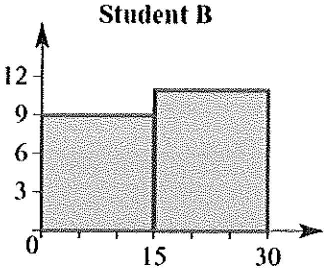
**9. [3 marks: 1, 1, 1]**

Two students show different histograms for the same set of data.

**Student A**




**Student B**




**a)** Which histogram is more useful in helping to analyse the data? Justify your decision.

A - smaller interval range

½ - can see more information ½.

**b)** What would you advise student B to do differently when constructing the histogram?

smaller interval widths ✓

**~ END OF TEST SECTION 1 ~**

Name: ______________________ Date: ________

## SECTION 2: CALCULATOR ASSUMED

| | | | |
|---|---|---|---|
| **Time:** | *37 minutes* | **Marks for Section 2:** | *34* |
| **Reading:** | *3 minutes* | **Equipment Allowed:** | *½ page notes (A4 one side), Scientific calculator* |
| **Working:** | *34 minutes* | | |

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**10. [3 marks: 1, 1, 1]**
Data was collected for a footy tipping competition one week, and the results are summarised below:

| *Score* ( *x* ) | *Frequency* ( *f* ) | *fx* |
|---|---|---|
| 6 | 4 | 24 |
| 7 | 9 | 63 |
| 8 | 10 | 80 |
| 9 | 7 | 63 |
| | $\Sigma f$ = 30 | $\Sigma fx$ = 230 |

**a)** What is the mode of the scores?

8 ✓

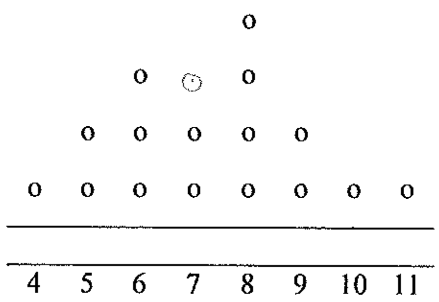
**b)** What is the median of the scores?

8 ✓

**c)** What is the mean of the scores (correct to one decimal place)?

7·67 ✓

**11. [1 mark]**
Heather records the ages of her 16 young cousins in the dot plot.




She realises that she has left out a cousin who is 7.

Which statistical measures (mean, median, mode, range) will change when she includes the extra score?

mean and median

½ ½

**12.** **[2 marks]**

The results on the same test for two classes are shown in the tables.

Ms Hypatia

| Score | Frequency |
|---|---|
| 20 | 0 |
| 21 | 1 |
| 22 | 8 |
| 23 | 7 |
| 24 | 7 |
| 25 | 6 |

Mr Kepler

| Score | Frequency |
|---|---|
| 20 | 3 |
| 21 | 6 |
| 22 | 7 |
| 23 | 8 |
| 24 | 3 |
| 25 | 2 |

Explain why comparing the mode for each class would not give an accurate comparison of the two class's performances on the test.

The mode is higher for Mr Kepler, but Ms Hypatia has 20 students with scores of 23 or better, compared to 13 for Mr Kepler ✓

The mean or median would be a better measure to compare ✓

**13.** **[4 marks: 2, 1, 1]**

Consider the following stem-and-leaf plot:

| STEM | LEAF |
|---|---|
| 0 | 6, 7 |
| 1 | 4, 7 |
| 2 | 1, 5, 3, 3, 5, 3 |
| 3 | 7, 4, 1, 5 |
| 4 | 6, 2 |

**a)** Redraw the stem-and-leaf plot as an ordered stem-and-leaf plot.

| Stem | Leaf |
|---|---|
| 0 | 6 7 |
| 1 | 4 7 |
| 2 | 1 3 3 3 5 5 |
| 3 | 1 4 5 7 |
| 4 | 2 6 |

✓✓

key 1|1 = 11.

**b)** Calculate the median

$$\frac{23+25}{2} = 24$$ ✓

**c)** Calculate the mean.

$\bar{x} = 25.5625$ ✓

**14. [8 marks: 2, 3, 1, 2]**
The following frequency table shows the distance thrown, in metres, by the students in the javelin competition at the Annual Athletics Carnival.

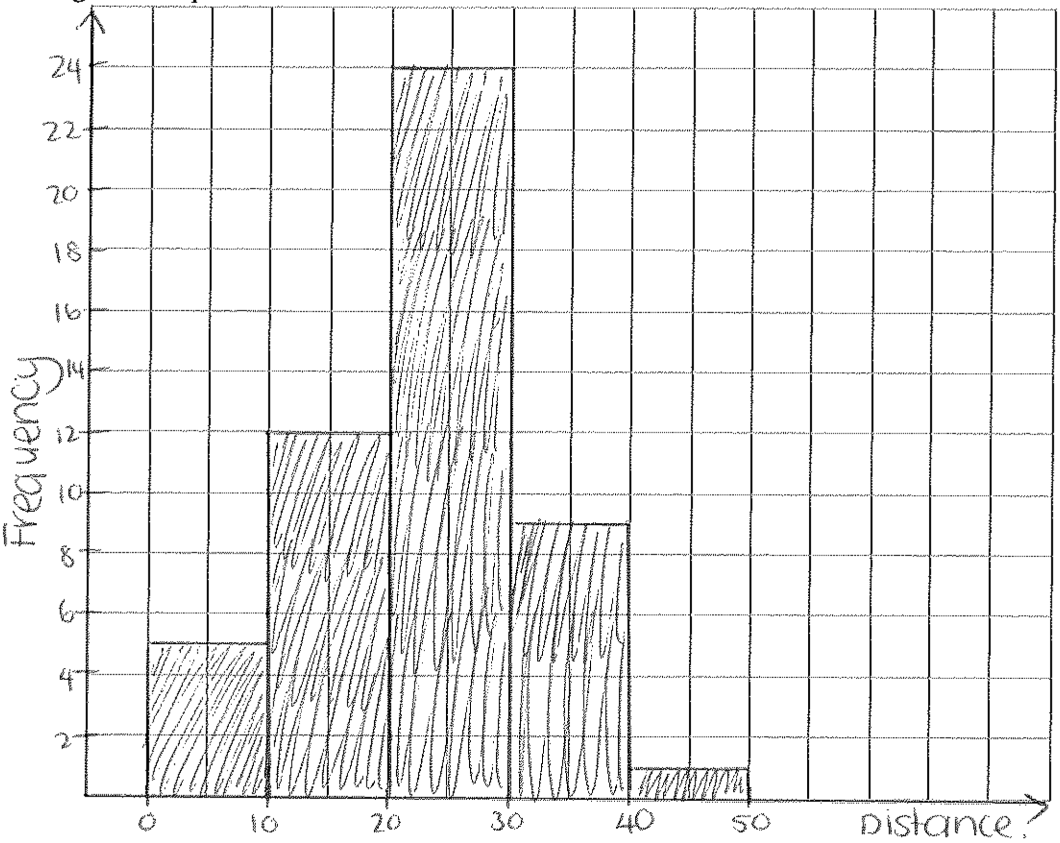
| Distance (m) | Frequency | fx |
|---|---|---|
| 0–<10 5 | 4 | 20 |
| 10–<20 15 | 12 | 180 |
| 20–<30 25 | 24 | 600 |
| 30–<40 35 | 9 | 315 |
| 40–<50 45 | 1 | 45 |
| | 50 | 1160 |

**a)** Calculate an estimate for the mean distance thrown by the competitors.

$\bar{x} = 23.2$ ✓✓

**b)** Draw a histogram to represent the data set.

✓ labels
✓ columns
✓ doesn't start on y axis
$-\frac{1}{2}$ gaps.




**c)** Use the shape of the histogram to describe the data set.

The large number of values in the middle band makes this data relatively symmetrical ✓
Slight positive skew.

**d)** 15 students were disqualified without recording a valid distance. Would it be reasonable to include these results? If you did record them what effect would they have on the mean and the median value?

As it was not a legal throw, the values should not be included ✓
If they were included, they should be recorded as 0. This would decrease the mean, but would not affect the median interval. ✓

**15.** **[11 marks: 5, 4, 1, 1]**

The following lists are of heights, in cm, of 30 Year 9 students from each of NSW and Victoria recorded in 2011.

NSW students:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 164 | 160 | 167 | 164 | 176 | 156 | 188 | 164 | 166 | 174 | 179 | 167 |
| 142 | 148 | 175 | ~~126~~ | 155 | 150 | 173 | 179 | 146 | 182 | 152 | 160 |
| 170 | 172 | 162 | 174 | 182 | 152 | | | | | | |

Victoria students:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 167 | 164 | 166 | 186 | 167 | 165 | 182 | 161 | 177 | 157 | 161 | 162 |
| 165 | 165 | 172 | 166 | 167 | 178 | 160 | 160 | 167 | 174 | 175 | 164 |
| 167 | 175 | 159 | 168 | 162 | 174 | | | | | | |

**a)** Draw a back-to-back stem-and-leaf plot of the data.

| NSW Leaf | Stem | Victoria - Leaf |
|---|---|---|
| 6 | 12 | |
| | 13 | |
| 8 6 2 | 14 | |
| 6 5 2 2 0 | 15 | 7 9 |
| 6 7 7 4 4 4 2 0 0 | 16 | 0 0 1 1 2 2 4 4 5 5 5 6 6 7 7 7 7 7 8 |
| 9 9 6 5 4 4 3 2 0 | 17 | 2 4 4 5 5 7 8 |
| 8 2 2 | 18 | 2 6 |

-1 if not ordered

✓✓ ✓ ✓✓

**b)** Find the mean and median height for the students from each State. Give answers correct to one decimal place if necessary.

NSW:
$\bar{x} = 164.2$ ✓
median = 165 ✓

Victoria:
$\bar{x} = 167.8$ ✓
median = 166.5 ✓

**c)** What conclusions can you draw from the stem plot and the statistics calculated regarding the height of Year 9 students from NSW compared to those from Victoria?

Although the mean and medians are close, the range of heights is quite different ½.
The victorian data is clustered around the 160-169 interval, whereas NSW is more spread out ½.

**d)** The mean height of Year 9 students in 2011 was reported as 165.38 cm and the median as 167.0 cm. How do you think the samples above compare to these values? Give some detail in your answer.

The values obtained in the samples are quite close to the values stated for the population. ✓ Some deviation is expected.

**16.** **[5 marks]**
Compare the following data sets using as many statistics as you can.

Set A: 1, 2, 4, 7, 11, 12, 14, 16 Set B: 3, 5, 6, 7, 10, 11, 12, 13

Set A:

$\bar{x} = 8.375$

median = 9

mode = none

range = 15

✓✓

Set B:

$\bar{x} = 8.375$

median = 8.5

mode = none

range = 10

✓✓

The greater range of set A has forced its median to be greater ✓

or set A has a larger spread of data than set B.

~ **END OF TEST SECTION 2** ~